

Performance Testing

Date	27 September 2026
Team ID	SWTID-2026-3833
Project Name	AI StudyBuddy API
Maximum Marks	4 Marks

Performance Testing - Benchmarked Sample Calls

The four calls below were captured directly from Postman/Thunder Client against the team's live MongoDB Atlas cluster, with response time and payload size read from the client for each call.

S.No	Endpoint	Method	Status	Response Time	Response Size
1	/api/auth/register (sample 1 - user "selva")	POST	201 Created	752 ms	543 B
2	/api/auth/register (sample 2 - user "bharathi")	POST	201 Created	529 ms	546 B
3	/api/auth/logout	POST	200 OK	8 ms	24 B
4	/api/materials/upload	POST	201 Created	124 ms	4.13 KB

Observations

- Registration (~530-750 ms) is the slowest MongoDB-only call because it performs a bcryptjs hash-and-salt round on every request before writing the new user document to Atlas.
- Logout is near-instant (8 ms) since it only issues a confirmation response with no database write.
- Material upload (124 ms) includes multer's disk write and file-type/size validation in addition to the MongoDB Atlas insert, and stays well within an acceptable range for a 5 MB upload ceiling.
- Round-trip time to the team's MongoDB Atlas cluster (rather than a local database) is a contributing factor to all four measured calls, compared with a purely local-MongoDB setup.

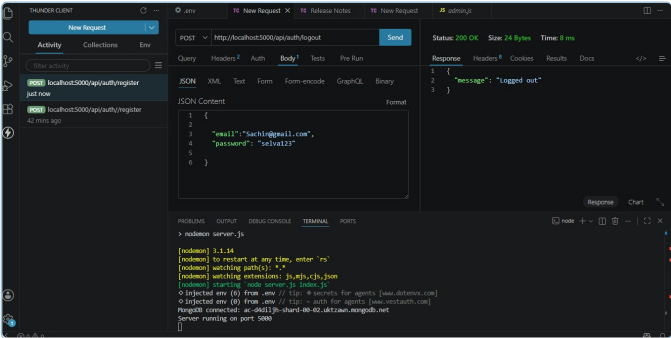
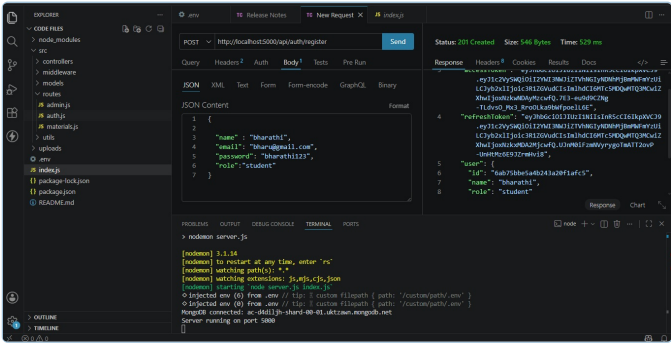
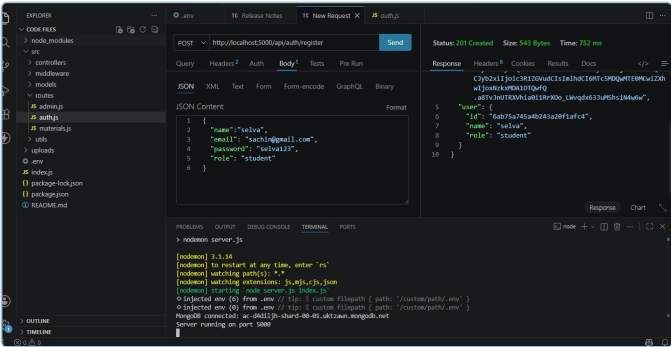
Other Implemented Endpoints (Functionally Verified)

The remaining endpoints below were exercised and confirmed working via the shared Postman/Thunder Client collection during Epic 5, but were not captured with an individual timing screenshot the way the four calls above were.

Endpoint	Method	Purpose
/api/auth/login	POST	Authenticates a registered user and issues a fresh access/refresh token pair
/api/materials/id/summarize	POST	Generates an AI summary of the material via Google Gemini
/api/materials/id/flashcards	POST	Generates AI flashcards (question/answer) from the material
/api/materials/id/quiz	POST	Generates an AI multiple-choice quiz from the material
/api/materials/id/study-plan	POST	Generates a personalized AI study plan from the material
/api/admin/*	GET/PUT/DELETE	Administrator-only user and system management endpoints

The four AI-generation calls above are expected to run noticeably slower than the MongoDB-only calls in the benchmark table, since each depends on a live round trip to the external Google Gemini API rather than only the Express/MongoDB layer.

Evidence - Sample Test Executions





AI StudyBuddy · Team SWTID-2026-3833 · Generated as part of Project Documentation (15–29 September 2026)